How a PWR Works

The Pressurised Water Reactor — structure & mechanisms

A plain-language companion to the NEXUS-1 console

What this is	A walk through the machine the NEXUS-1 console simulates: what the major parts are, how water carries heat out of the reactor, and how that heat becomes electricity — explained for a curious non-specialist.
Modelled on	The same reactor the demo drives: a two-loop PWR with two U-tube steam generators (SG-1/SG-2), two reactor-coolant pumps (RCP-1A/1B), a pressuriser, and a 157-assembly core with RCCA control-rod banks.
Diagrams	Four 2-D schematics — the whole plant, the reactor vessel, the steam generator, and the fuel & core. Simplified and *not to scale*; no dimensions.
Build	NX1-094C79E0A366D34F
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How to read these schematics. They show *how the parts connect and how heat moves*, not how big anything is. Shapes are stylised, nothing is to scale, and there are no dimensions — the goal is a clear mental picture of the structure and the mechanism, matching the system the console models.

1 The idea in one minute

A pressurised water reactor makes electricity the same way a coal or gas plant does — by boiling water into steam and using that steam to spin a turbine. The only thing that is different is the *heat source*: instead of a furnace, a controlled nuclear chain reaction in the reactor core heats the water.

The defining trick of a PWR is in its name. The water that flows through the radioactive core is kept under very high pressure — around 155 times atmospheric — so that even at roughly 315 °C it *cannot boil*. It stays liquid. That super-hot, pressurised water is pumped to a device called a **steam generator**, where it gives up its heat to a completely separate batch of water, boils *that* water into steam, and returns to the core to be reheated. The two bodies of water never touch.

That separation is the whole safety idea: the water that becomes steam, drives the turbine, and travels all over the plant is *clean*; the water that has been through the reactor is *sealed* inside a closed, shielded loop. Figure 1 shows the two loops side by side.

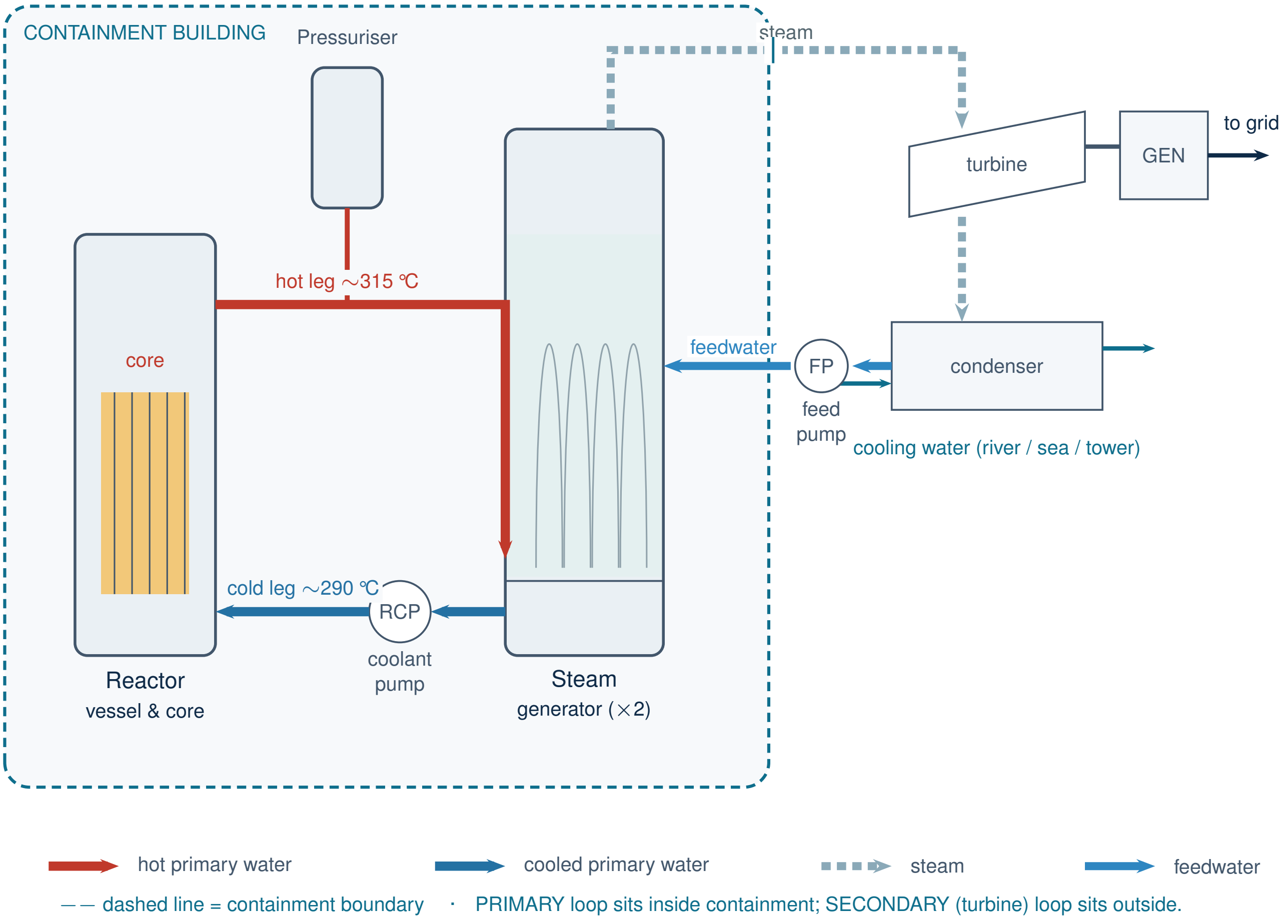

Figure 1: The whole plant. **Primary loop** (red/blue, sealed inside containment): the core heats pressurised water; a pump drives it through the steam generator and back. **Secondary loop** (steam/feedwater, outside): the steam generator boils clean water, the steam spins the turbine, the condenser turns it back to water. The two loops exchange heat but never mix.

One action, everywhere. This connected picture is exactly what the console animates: move the control rods and the change flows core → coolant → steam generator → turbine → grid in real time, just as the arrows above suggest.

2 Following the water: the two loops

2.1 The primary loop — sealed, hot, and under pressure

Start at the reactor core (left of Figure 1). Water leaves the top of the vessel through a thick pipe called the **hot leg** at around 315 °C. Because the whole loop is held near 155 bar by the **pressuriser**, this water stays liquid instead of flashing to steam. The hot leg carries it to the **steam generator**, where it gives up most of its heat, and leaves a little cooler (around 290 °C) through the **cold leg**. A **reactor coolant pump** then pushes it back into the vessel to be reheated. Our reference plant has two such loops running in parallel — two hot legs, two steam generators, two pumps — which is why the diagrams mark the steam generator "$\times 2$".

The pressuriser deserves a word. It is a tall tank, partly water and partly steam, connected to one hot leg. By heating or spraying its contents it nudges the pressure of the entire primary loop up or down — like the expansion tank that keeps a car's cooling system at the right pressure. Keeping the primary water just below boiling, everywhere, is its single job.

2.2 The secondary loop — clean steam to the turbine

Now follow the steam side (right of Figure 1). Inside the steam generator, separate "feedwater" boils into steam. That steam leaves the top, crosses out of the containment building, and spins the **turbine**, which turns the **generator** that makes electricity. Spent steam falls into the **condenser**, where a third stream of cooling water (from a river, the sea, or a cooling tower) chills it back into liquid. A **feed pump** returns that water to the steam generator, and the cycle repeats. This secondary water is never radioactive — which is why the turbine hall can sit outside the thick containment shell.

3 Inside the reactor vessel

The reactor vessel is a thick steel pot, sealed by a bolted-on head, that holds the core and steers the coolant through it. Figure 2 cuts it open.

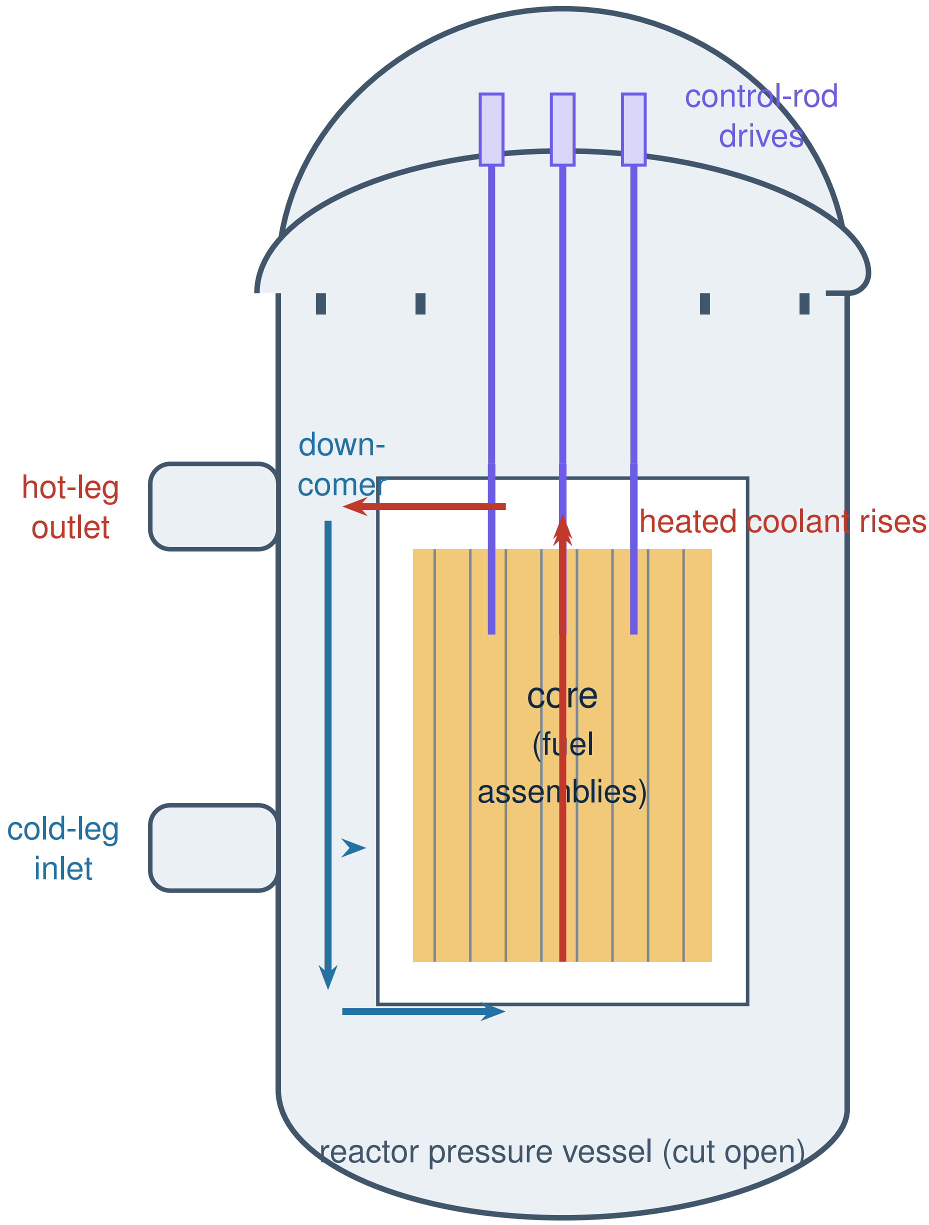

Figure 2: Reactor vessel, cut open. Cool water enters the side (**cold-leg inlet**), is steered *downward* through the gap around the core — the **downcomer** — then turns and flows *up* through the fuel, heating as it goes, and leaves through the **hot-leg outlet**. **Control rods** (violet) hang from drive mechanisms on the head and drop into the core to reduce or stop the reaction.

3.1 How the power is steered — and stopped

Two mechanisms set how hard the core runs. **Control rods** — bundles of neutron-absorbing material — slide in and out of the fuel from above; pushing them in soaks up neutrons and lowers power, pulling them out raises it. For slow, even adjustments over weeks, operators also dissolve **boron** (another neutron absorber) in the coolant and gradually change its concentration. The console exposes both: the rod banks on the *Control Rods* page and boron concentration on the *Core* page.

Shutdown is the same mechanism taken to its limit. In a **SCRAM**, the rods are released and fall fully into the core under gravity in seconds, stopping the chain reaction almost at once. "Almost" matters: a small fraction of neutrons arrive late (the "delayed" neutrons), so power tails off over seconds-to-minutes rather than vanishing instantly — a behaviour the demo's physics engine reproduces faithfully.

4 The steam generator — where the two worlds meet

This is the component that makes a PWR a PWR: the wall where heat crosses from the radioactive loop to the clean one *without the two waters ever mixing*. Figure 3 opens one up.

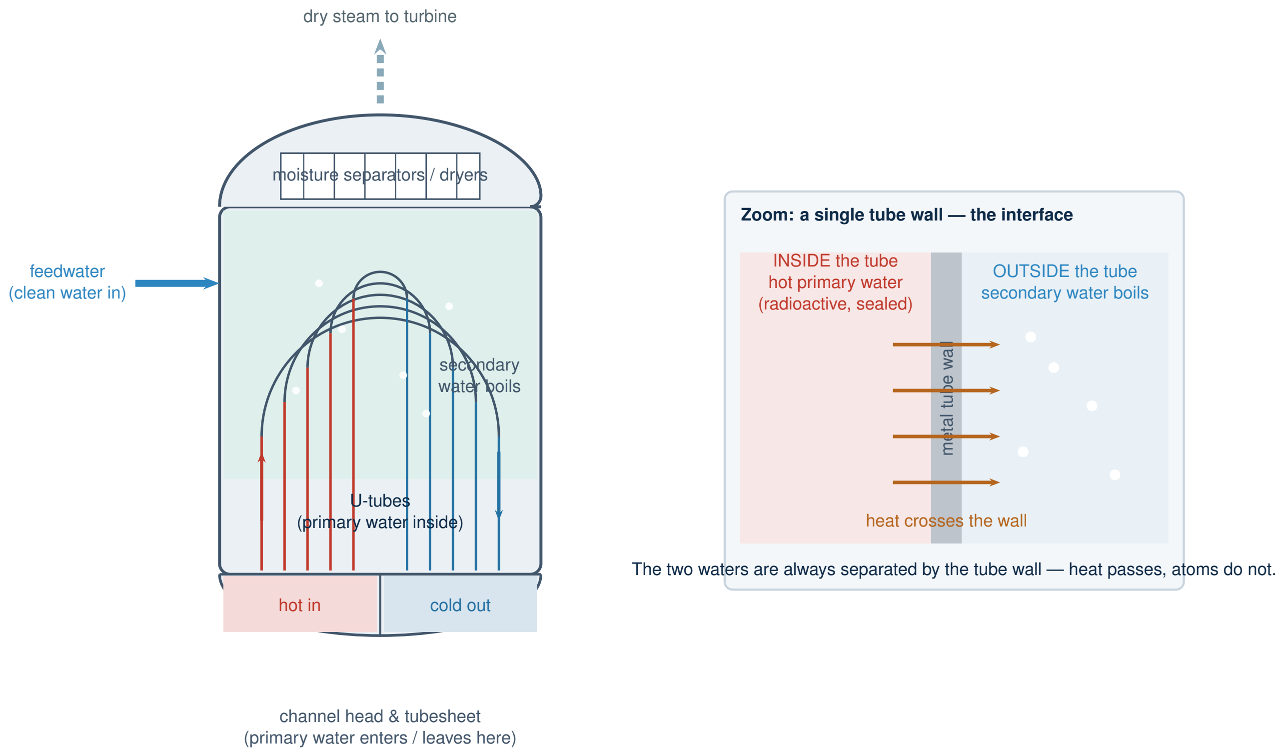

Figure 3: Steam generator, cut open (left) with the key idea enlarged (right). Hot primary water enters one half of the bottom **channel head**, travels up and over through thousands of **U-tubes**, and exits the other half — cooler — back to the cold leg. Clean **feedwater** fills the shell *around* those tubes, picks up their heat, and boils. **Dryers** at the top strip out water droplets so only dry steam reaches the turbine. The enlargement shows why a PWR is safe by design: heat conducts through each tube wall, but the radioactive and clean waters stay on opposite sides of it.

Why this barrier matters. The tube wall is one of the plant's defences that keeps radioactivity penned inside the primary loop. It is also why the digital twin watches the gap between "measured" and "modelled" so closely: a slow, growing divergence on a steam-generator signal is exactly the kind of early warning a real plant looks for.

5 The fuel and the core

The heat all comes from one place: the fuel. Figure 4 zooms from the whole core down to a single fuel pellet.

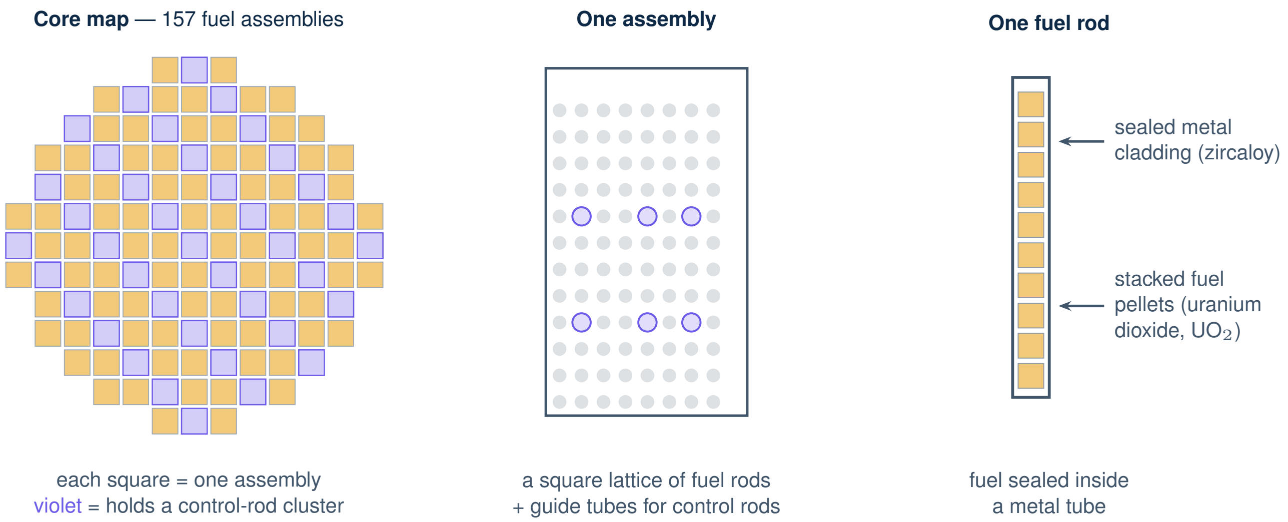

Figure 4: From core to pellet. The **core** is a near-circular array of 157 **fuel assemblies**; some positions hold clusters of control rods (violet). Each **assembly** is a square lattice of fuel rods threaded with empty guide tubes the control rods slide into. Each **fuel rod** is a sealed metal tube packed with ceramic **uranium-dioxide pellets** — the actual fuel.

5.1 The chain reaction, briefly

A uranium nucleus that absorbs a neutron can **fission** — split in two, release heat, and throw out two or three fresh neutrons. If on average one of those goes on to split another nucleus, the reaction is steady; the core is “critical” (the normal operating state, not a fault). The surrounding water does double duty here: it carries the heat away *and* slows the neutrons down to the speed at which they cause fission best — it is both coolant and “moderator”.

That second role gives a PWR a built-in safety feature. If the water heats up or boils, it thins out and moderates less effectively, so the reaction naturally eases off. Power that runs high tends to pull *itself* back down. This negative feedback — together with the Doppler effect in the fuel — is what makes the reactor stable, and it is precisely the behaviour the console’s point-kinetics engine integrates live.

6 Where each part lives in the console

Every component above has a home screen in the demo. This is the bridge between the machine and the simulation.

Real component	Console screen
Core, fuel, chain reaction	*Reactor Kinetics* (the live physics), *Core* (parameters, core map)
Control rods & boron	*Control Rods* (rod banks, SCRAM), *Core* (boron)
Reactor vessel & coolant flow	*Coolant / TH*, *3D Reactor View*
Steam generators	*Steam & Secondary*, *Coolant / TH*
Pressuriser & primary pressure	*Coolant / TH*
Turbine, generator, condenser	*Steam & Secondary*, *Power & Grid*
Whole-plant heat-to-grid flow	*Plant Overview*, *Plant 3D View*, *Energy Flow*
Measured-vs-modelled signals	*Digital Twin*
Component condition & ageing	*Component Registry*, *Aging & Degradation*

A fair reminder. These schematics and figures are simplified teaching pictures of a representative two-loop PWR — stylised, not to scale, and free of dimensions on purpose. They capture how the parts connect and how heat moves, which is what the NEXUS-1 console models; they are not engineering drawings of any specific reactor.